

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1336

Henry Creek 138 kV
January 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1336. The Interconnection Customer proposes a 279.7 MW solar photovoltaic facility interconnecting at the Henry Creek 138 kV substation in the Kentucky area. The request was received on November 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 325 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,967,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ubc S Campus – Henry Crk 138 kV	Central Campus Unm – Henry Crk 138 kV	148.6	182	100.0
Ubc S Campus – Danz Ae Sw Sta 138 kV	Central Campus Unm – Henry Crk 138 kV L	137.1	192	100.0
Danz Ave Sw Sta – Cwec Tigerton Substation 138 kV L	Central Campus Unm – Henry Crk Substation 138 kV	126.2	202	100.0
Central Campus Unm – Henry Crk 138 kV	Ubc S Campus – Henry Creek 138 kV	121.1	223	100.0
Ellendae – Cwec Tigerton 138 kV	Central Campus Unm – Henry Crk Substation 138 kV	118.4	209	100.0
Albion Rd – Central Campus Unm 138 kV	Ubc S Campus – Henry Crk 138 kV line	110.9	234	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Henry Crk Substation – Ubc South Campus 138 kV L	Rebuild line with higher-capacity conductor	\$6,883,000	100% (\$6,883,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ubc S Campus – Danz Ae Sw Sta 138 kV	Rebuild line with higher-capacity conductor	\$15,856,000	100% (\$15,856,000)
Cwec Tigerton – Danz Ave Sw Sta 138 kV L	Rebuild line with higher-capacity conductor	\$6,159,000	100% (\$6,159,000)
Central Campus Unm – Henry Crk Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,256,000	100% (\$9,256,000)
Ellendae – Cwec Tigerton 138 kV	Rebuild line with higher-capacity conductor	\$4,929,000	100% (\$4,929,000)
Albion Rd – Central Campus Unm 138 kV	Rebuild line with higher-capacity conductor	\$5,884,000	100% (\$5,884,000)
	Total allocated to project		\$48,967,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.